

EECS 498-005 Project Proposal

From Game Theory to Machine Learning: Experiments on Reducing Overfitting



1 Motivation

Overfitting occurs when the model is too complex or there are too few samples, which causes the model to wrongly associate the feature with the labels and make incorrect predictions (Simpson et al., 2019). Overfitting is problematic because it restrains the model's ability to generalize well to new data. In order to maintain the model's performance on new dataset, we propose a new way inspired by the game theory to reduce overfitting. Given a pre-trained model $v(\cdot)$ and an input sample x , where $x \in \mathbb{R}^n$, we use $N = \{1, 2, \dots, n\}$ to denote the set of all input variables, and define $v(N)$ as the scalar output score of the model given the entire input N . Similarly, we can define $v(S)$ as the scalar output score of the model given only a subset of the input variables $S \subseteq N$. According to Harsanyi (1963), we can decompose the output score $v(N)$ into the sum of interaction utilities with respect to all possible subsets of the set of input variables:

$$v(N) = \sum_{S \subseteq N} I(S),$$

where $I(S)$ is recursively defined as

$$I(S) \stackrel{\text{def}}{=} v(S) - v(\emptyset) - \sum_{L \subsetneq S, |L| \geq 1} I(L).$$

Intuitively, $I(S)$ quantifies the interaction utility that is only triggered by the all the variables in S , but not triggered by any smaller subsets of S . It is reasonable to assume that the $I(S)$ where $|S|$ is large typically contains the information that lead to overfitting, because this interaction utility is only triggered by the collaboration of massive input variables (specifically, pixels in image classification). In the extreme case, if a model rely on all pixels of an image to do the classification, then it is terribly overfitted to the training data. Therefore, to reduce overfitting, we want to suppress the utilities

of those $I(S)$ with large $|S|$. In the following sections, we call the $I(S)$ with large $|S|$ *high-order*.

2 Supporting Materials

The models we will use are AlexNet and VGG16. They will be trained on the CIFAR-10 dataset. AlexNet (Krizhevsky, 2014) is a typical convolutional neural network architecture. Its basic building blocks are convolutions, max pooling and dense layers. VGG16 (Simonyan and Zisserman, 2014) is a convolutional neural network model for classification and detection. The CIFAR-10 dataset (Krizhevsky et al., 2009) contains 60000 32x32 colour images in 10 classes, with 6000 images per class.

3 Plans

We aim to design a new loss function to penalize the high-order $I(S)$. We will train two models on the image classification task with the original and our novel loss function to observe the effect of reducing overfitting. We plan to train the AlexNet and VGG16 on CIFAR-10. First, we will select a subset of the images in CIFAR-10 in order to magnify overfitting. Particularly, we will only use 10% of the training data and our selected training dataset will still be balanced, which consists of 5000 32x32 colour images in 10 classes, with 500 images per class. Next, we will use the original loss function without penalizing the high-order $I(S)$ and train AlexNet and VGG16 for the same number of epochs. We will generate the accuracy and loss plots. Then, we will implement our new loss function and train AlexNet and VGG16 under the new loss function. We will also plot the accuracy and loss graphs in order to make comparisons. Ideally, by comparing the two sets of plots, we will observe that the overfitting phenomenon will be alleviated under the new loss function. Also, we will observe an increase in test accuracy due to less overfitting.

References

- John C. Harsanyi. 1963. [A simplified bargaining model for the n-person cooperative game](#). *International Economic Review*, 4(2):194–220.
- Alex Krizhevsky. 2014. [One weird trick for parallelizing convolutional neural networks](#). *CoRR*, abs/1404.5997.
- Alex Krizhevsky, Geoffrey Hinton, et al. 2009. Learning multiple layers of features from tiny images.
- Karen Simonyan and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.
- Becks Simpson, Francis Dutil, Yoshua Bengio, and Joseph Paul Cohen. 2019. Gradmask: Reduce overfitting by regularizing saliency. *arXiv preprint arXiv:1904.07478*.